MIE Integral Quiz

$$\int_0^4 \int_0^{\sqrt{y}} xy^2 \, dx \, dy$$

$$= \int_0^4 y^2 \int_0^{\sqrt{y}} x \, dx \, dy$$

$$= \int_0^4 y^2 \left[\tfrac{1}{2}x^2\right]_0^{\sqrt{y}} dy$$

$$= \int_0^4 y^2 \left[\tfrac{1}{2}(\sqrt{y})^2 - \tfrac{1}{2}(0)^2\right] dy$$

$$= \int_0^4 \tfrac{1}{2} y^3 \, dy$$

$$= \left[\tfrac{1}{8} y^4\right]_0^4$$

$$= \tfrac{1}{8}(4)^4 - \tfrac{1}{8}(0)^4$$

$$= \tfrac{1}{8}(256)$$

$$= \boxed{32}$$

$$\int_0^1 \int_y^{2y} \int_0^{x+y} 6xy \, dz \, dx \, dy$$

$$= \int_0^1 \int_y^{2y} 6xy \left[z\right]_0^{x+y} dz \, dy$$

$$= \int_0^1 \int_y^{2y} 6(x^2y + xy^2) \, dx \, dy$$

$$= \int_0^1 6\left[\frac{x^3}{3}y + \frac{x^2}{2}y^2\right]_y^{2y} dy$$

$$= \int_0^1 \left[2x^3y + 3x^2y^2\right]_y^{2y} dy$$

$$= \int_0^1 \left[2(8y^3)y + 3(4y^2)y^2 - 2(y^3)y - 3(y^2)y^2\right] dy$$

$$= \int_0^1 \left[16y^4 + 12y^4 - 2y^4 - 3y^4\right] dy$$

$$= \int_0^1 23y^4 \, dy$$

$$= \left[\tfrac{23}{5}y^5\right]_0^1$$

$$= \tfrac{23}{5}(1)^5 - \tfrac{23}{5}(0)^5$$

$$= \boxed{\frac{23}{5}}$$

$$\int_1^3 \int_{-1}^2 \int_{-y}^z \frac{z}{y}\, dx\, dz\, dy$$

$$= \int_1^3 \int_{-1}^2 \frac{z}{y} [x]_{-y}^{z}\, dz\, dy$$

$$= \int_1^3 \int_{-1}^2 \frac{z}{y} (z+y)\, dz\, dy$$

$$= \int_1^3 \int_{-1}^2 \left(\frac{z^2}{y} + z\right) dz\, dy$$

$$= \int_1^3 \left[\frac{z^3}{3y} + \frac{z^2}{2}\right]_{-1}^{2} dy$$

$$= \int_1^3 \left[\frac{8}{3y} + \frac{4}{2} - \frac{(-1)}{3y} - \frac{(-1)^2}{2}\right] dy$$

$$= \int_1^3 \left[\frac{9}{3y} + \frac{3}{2}\right] dy$$

$$= \int_1^3 \left[\frac{3}{y} + \frac{3}{2}\right] dy$$

$$= \left[3 \ln|y| + \frac{3}{2} y\right]_1^3$$

$$= 3 \ln 3 + \frac{3}{2}(3) - 3 \ln 1 - \frac{3}{2}$$

$$= 3 \ln 3 + \frac{9}{2} - \frac{3}{2}$$

$$= 3 \ln 3 + \frac{6}{2}$$

$$= \boxed{3 \ln 3 + 3}$$

$$\int_0^1 \int_{2x}^2 (x-y)\, dy\, dx$$

$$= \int_0^1 \left[xy - \frac{y^2}{2}\right]_{2x}^{2} dx$$

$$= \int_0^1 \left[2x - \frac{4}{2} - 2x^2 + \frac{4x^2}{2}\right] dx$$

$$= \int_0^1 [2x - 2]\, dx$$

$$= [x^2 - 2x]_0^1$$

$$= (1)^2 - 2(1) - 0$$

$$= \boxed{-1}$$

$$\int_0^2 \int_0^{z^2} \int_0^{y-z} (2x - y)\, dx\, dy\, dz$$

$$= \int_0^2 \int_0^{z^2} \left[x^2 - xy \right]_0^{y-z} dy\, dz$$

$$= \int_0^2 \int_0^{z^2} \left[(y-z)^2 - (y-z)y \right] dy\, dz$$

$$= \int_0^2 \int_0^{z^2} \left(y^2 - 2yz + z^2 - y^2 + yz \right) dy\, dz$$

$$= \int_0^2 \int_0^{z^2} \left[-yz + z^2 \right] dy\, dz$$

$$= \int_0^2 \left[-\frac{y^2}{2} z + yz^2 \right]_0^{z^2} dz$$

$$= \int_0^2 \left[-\frac{z^4}{2} z + z^2 z^2 \right] dz$$

$$= \int_0^2 \left[-\frac{z^5}{2} + z^4 \right] dz$$

$$= \left[-\frac{z^6}{12} + \frac{z^5}{5} \right]_0^2$$

$$= \left[-\frac{(2)^6}{12} + \frac{(2)^5}{5} \right]$$

$$= -\frac{64}{12} + \frac{32}{5}$$

$$= -\frac{16}{3} + \frac{32}{5}$$

$$= \boxed{\frac{16}{15}}$$